AWS Glue Customer Orders ETL Pipeline

Project report | Completed October 20, 2026 | AWS Region: us-east-1

An end-to-end data engineering lab using Amazon RDS MySQL, AWS Glue with PySpark, Amazon S3 Parquet, and Amazon Athena.

8	$2,159.93	Successful
Validated orders	Total sales	Athena query

1. Executive summary

Designed, built, debugged, and validated a cloud ETL pipeline. AWS Glue reads customer and order data from RDS MySQL through JDBC, performs a customer-to-order inner join, derives a customer name and an order total, and writes compressed Parquet output to S3. Amazon Athena queries the resulting dataset. A completed Athena aggregation returned **8 orders** and **$2,159.93** in sales.

2. Solution architecture

SOURCE	Amazon RDS MySQL: glue_etl_demo.customers + orders
EXTRACT	Glue JDBC connection + Secrets Manager credentials
TRANSFORM	AWS Glue 5.1 / PySpark: join, customer_name, order_total
LOAD	Amazon S3: processed/customer_orders/ (Parquet)
QUERY	Athena: glue_etl_analytics.customer_orders

3. Configuration inventory

• Source database: glue_etl_demo; tables: customers, orders.

• Glue job: customer-orders-etl; Glue 5.1; 2 G.1X workers.

• Glue connection: glue-etl-mysql-connection; secret: glue-etl/mysql-credentials.

• Execution role: AWSGlueServiceRole; VPC: mysql-lab-vpc.

• Network: S3 gateway endpoint; Secrets Manager interface endpoint.

• S3 output: s3://glue-etl-lab-235494792645/processed/customer_orders/.

• Athena database: glue_etl_analytics; table: customer_orders.

4. Transformation logic

The Glue job performs an inner join on customer_id, concatenates first_name and last_name into customer_name, and calculates order_total as quantity multiplied by unit_price (rounded to two decimal places).

```
joined = customers.join(orders,
    customers.customer_id == orders.customer_id, "inner")
result = joined.select(
    customers.customer_id.alias("customer_id"),
    concat_ws(" ", customers.first_name,
              customers.last_name).alias("customer_name"),
    customers.email, customers.city, customers.state,
    orders.order_id, orders.order_date,
    orders.product_name, orders.quantity,
    orders.unit_price,
    round(col("quantity") * col("unit_price"), 2)
        .alias("order_total")
)
output_path = "s3://glue-etl-lab-235494792645/processed/customer_orders/"
result.write.mode("overwrite").parquet(output_path)
```

This is the verified transformation and output excerpt, not a complete standalone Glue script.

5. Athena schema and validation

Athena table customer_orders exposes 11 fields: customer_id (int), customer_name (string), email (string), city (string), state (string), order_id (int), order_date (date), product_name (string), quantity (int), unit_price (decimal(10,2)), and order_total (decimal(13,2)).

```
SELECT COUNT(*) AS total_orders,
       ROUND(SUM(order_total), 2) AS total_sales
FROM glue_etl_analytics.customer_orders;

-- Observed result: total_orders = 8,
--                  total_sales = 2159.93
```

6. Implementation and troubleshooting

• Created RDS MySQL source tables and sample records; tested the Glue JDBC connection.

• Configured the Glue execution role to retrieve the secret from Secrets Manager.

• Created private VPC service endpoints for S3 and Secrets Manager access.

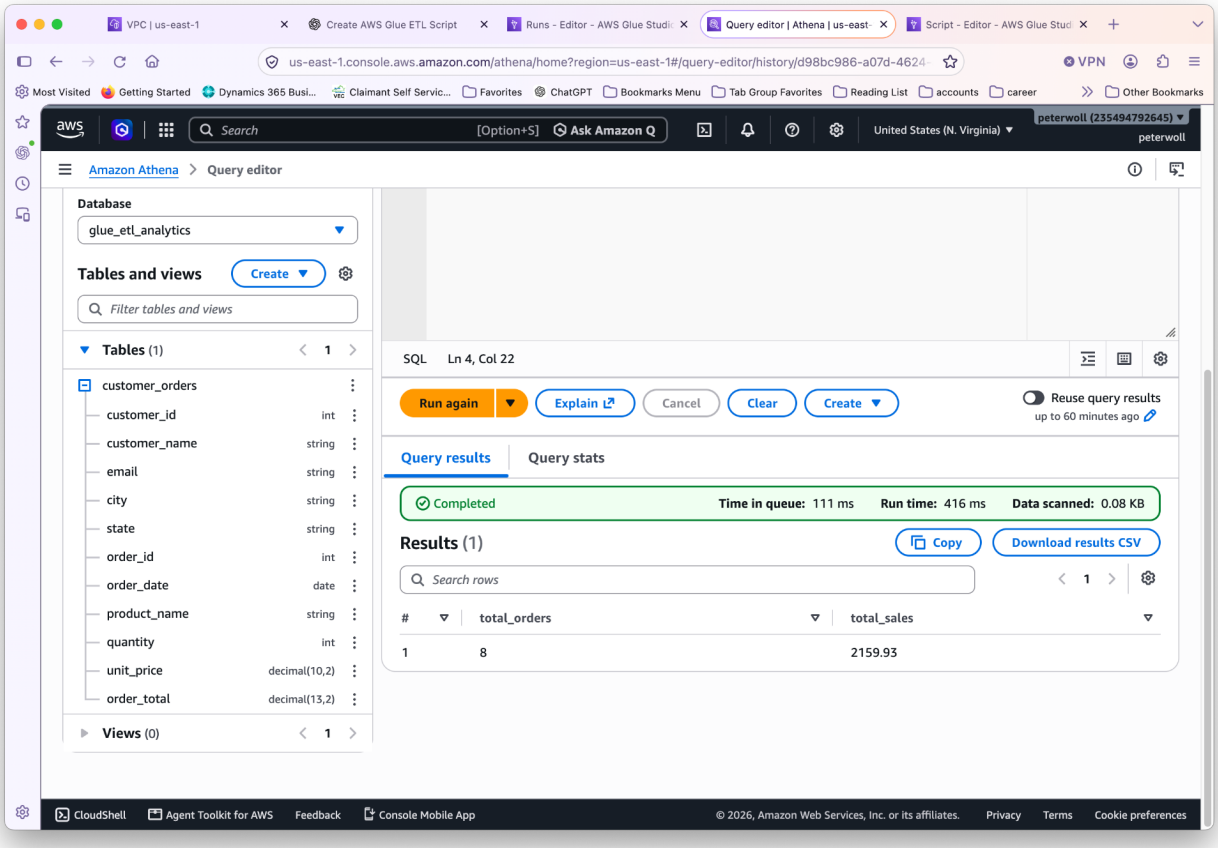
• Built the PySpark job and selected Parquet as the output format.

• Resolved an S3 AccessDenied error: the initial role policy referenced the wrong output bucket; corrected it to glue-etl-lab-235494792645.

• Reran the Glue job successfully and confirmed the Parquet file in S3.

• Created the Athena external table and verified the output using SELECT and aggregate SQL.

7. Evidence: Athena query results



Athena screenshot showing a completed query with 8 orders and 2159.93 total sales. Contains account identifiers; redact before public distribution.

8. Security, cost and portfolio notes

Use Secrets Manager for database credentials, least-privilege IAM policies, and security-group-restricted database access. Review ongoing costs for RDS, Glue, interface endpoints, S3 and Athena. Preserve sanitized evidence before deleting lab resources. Do not upload passwords, secrets, or live connection strings to GitHub.

Portfolio statement: Built and validated an AWS ETL pipeline using RDS MySQL, Glue/PySpark, S3 Parquet and Athena; transformed customer-order data and verified 8 orders totaling $2,159.93.

Companion deliverable: README_AWS_Glue_Customer_Orders_ETL.md